Referee package (DRAFT) – RES-4 closure: the STEP-5B layer-closure ratio is positive and monotone across the intensity band

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

This referee document concerns RES-4 (the H-LAYER-AUX folder): the positivity and monotonicity of the STEP-5B layer-closure ratio across the intensity band. **Scope:** the layer-closure ratio $\rho(I) = K_b(I)/K(I)$ on $I \in [4 \times 10^{-4}, 2 \times 10^{-3}]$ at the anchor. **Not claimed:** this is the layer-ratio positivity, an input to STEP-5B, not the full STEP-5B closure.

2. The result

The STEP-5B layer-closure ratio $\rho(I) = K_b(I)/K(I)$ is strictly monotone decreasing ($\rho'(I) < 0$ at all 201 nodes) and satisfies $\rho(I) \geq \rho(2 \times 10^{-3}) = \times 2.58 > 1$ for all I in the band at the anchor. The layer therefore closes with a positive ratio across the whole intensity range, binding at the endpoint.

3. Structure of the argument

The budget $K_b(I)$ and consumption $K(I)$ are formed from the Gershgorin layer bound and the off-shell operator decision; their ratio is evaluated on a 201-node intensity grid, its derivative sign checked node-by-node, and the worst (endpoint) value certified > 1. Monotonicity plus the endpoint value give band-wide positivity.

4. Numerical reproduction

Three scripts (the Gershgorin layer bound, the RES-4 intensity sweep, the off-shell operator decision) reproduce $\rho'(I) < 0$ and $\rho \geq \times 2.58$; run them from the bundle.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer.

(β) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is REVIEW DRAFT, NOT operator-confirmed, and the claim-ledger tier is the arbiter.

(γ) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder's bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values it cites are reproduced by the folder's scripts (all asserts pass).

6. Result footer

```
Result ID:               B5-BEYOND-LAYER-BOUND / H-LAYER-AUX (RES-4)
                         referee package (DRAFT)
Precise statement:       The STEP-5B layer-closure ratio
                         rho(I)=K_b(I)/K(I) is strictly monotone
                         decreasing (rho'(I)<0 at all 201 nodes) and
                         rho(I) >= rho(2e-3) = x2.58 > 1 for all I in
                         [4e-4, 2e-3] at the anchor; the layer closes
                         with a positive ratio band-wide, binding at
                         the endpoint.
Scope:                   layer-closure ratio rho(I) on the intensity
                         band at the anchor. Input to STEP-5B.
Evidence grade:          intensity-band layer-ratio positivity (T5);
                         EXECUTED (3 scripts PASS).
Reproduction command:    python
                         codes/vacuum/beyond_layer_gershgorin_bound.py
                         ; hlayer_res4_intensity_sweep.py ;
                         res5_024_offshell_operator_decision.py
Falsifier:               an intensity in the band where rho(I)<=1, or
                         where rho'(I)>=0.
Tier before / after:     DRAFT referee package; RES-4 layer-ratio
                         positivity T5. NOT operator-confirmed.
No-overclaim statement:  the layer-ratio positivity across the band; an
                         input to STEP-5B, not its full closure.
Next required action:    operator review + confirm -> PUBLISHED bundle.
```